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CSCI596 Assignment 2—Message Passing Interface

Both global_avg.c and global_avg.out files are attached as part of this submission. The content of these files are also pasted below as well.

- 1. Submit the source code as well as the printout from a test run on 4 processors and that on 8 processors.**

global_avg.c:

```
#include "mpi.h"
#include <stdio.h>

int nprocs; // Number of processes
int myid;   // My rank

double global_sum(double partial) {
    /* Write your hypercube algorithm here */
    double mydone = partial;
    double hisdone;
    MPI_Status status;

    for (int bitvalue=1; bitvalue<nprocs; bitvalue*=2)
    {
        int partner = myid ^ bitvalue;
        //send mydone to partner;
        MPI_Send(&mydone, 1, MPI_DOUBLE, partner, bitvalue, MPI_COMM_WORLD);
        //receive hisdone from partner;
        MPI_Recv(&hisdone, 1, MPI_DOUBLE, partner, bitvalue, MPI_COMM_WORLD,
&status);
        mydone = mydone + hisdone;
    }
    return mydone;
}

int main(int argc, char *argv[]) {
    double partial, sum, avg;
    double cpu1, cpu2;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &myid);
    MPI_Comm_size(MPI_COMM_WORLD, &nprocs);
```

```

partial = (double) myid;
printf("Rank %d has partial value %le\n", myid, partial);

cpu1 = MPI_Wtime();
sum = global_sum(partial);
cpu2 = MPI_Wtime();

if (myid == 0) {
    avg = sum/nprocs;
    printf("Global average = %le\n", avg);
    printf("Execution time (s) = %le\n",cpu2-cpu1);
}

MPI_Finalize();
return 0;
}

```

global_avg.out:

```

Rank 0 has partial value 0.000000e+00
Rank 2 has partial value 2.000000e+00
Rank 1 has partial value 1.000000e+00
Rank 3 has partial value 3.000000e+00
Rank 7 has partial value 7.000000e+00
Rank 6 has partial value 6.000000e+00
Rank 4 has partial value 4.000000e+00
Rank 5 has partial value 5.000000e+00
Global average = 3.500000e+00
Execution time (s) = 1.178422e-02
Rank 0 has partial value 0.000000e+00
Rank 2 has partial value 2.000000e+00
Rank 3 has partial value 3.000000e+00
Rank 1 has partial value 1.000000e+00
Global average = 1.500000e+00
Execution time (s) = 1.265490e-04

```

```

joeyma@discovery2:~/cs596/as02
[joeyma@discovery2 as02]$ mpicc -o global_avg global_avg.c
[joeyma@discovery2 as02]$ vi global_avg.sl
[joeyma@discovery2 as02]$ sbatch global_avg.sl
Submitted batch job 11991993
[joeyma@discovery2 as02]$ squeue -u joeyma
      JOBID PARTITION   NAME     USER ST       TIME  NODES MODELIST(REASON)
    11991993      main global_a   joeyma R        0:18        2 b22-[02-03]
[joeyma@discovery2 as02]$ ls -lt
total 98
-rwxrwxr-x 1 joeyma joeyma 18384 Sep 14 12:49 global_avg
-rw-r--r-- 1 joeyma joeyma  1124 Sep 14 12:49 global_avg.c
-rw-r--r-- 1 joeyma joeyma   211 Sep 14 12:45 global_avg.sl
[joeyma@discovery2 as02]$ ls -lt
total 98
-rwxrwxr-x 1 joeyma joeyma 18384 Sep 14 12:50 global_avg
-rw-rw-r-- 1 joeyma joeyma     0 Sep 14 12:50 global_avg.out
-rw-r--r-- 1 joeyma joeyma  1124 Sep 14 12:49 global_avg.c
-rw-r--r-- 1 joeyma joeyma   211 Sep 14 12:45 global_avg.sl
[joeyma@discovery2 as02]$ cat global_avg.out
Rank 0 has partial value 0.000000e+00
Rank 2 has partial value 2.000000e+00
Rank 1 has partial value 1.000000e+00
Rank 3 has partial value 3.000000e+00
Rank 7 has partial value 7.000000e+00
Rank 6 has partial value 6.000000e+00
Rank 4 has partial value 4.000000e+00
Rank 5 has partial value 5.000000e+00
Global average = 3.500000e+00
Execution time (s) = 1.178422e-02
Rank 0 has partial value 0.000000e+00
Rank 2 has partial value 2.000000e+00
Rank 3 has partial value 3.000000e+00
Rank 1 has partial value 1.000000e+00
Global average = 1.500000e+00
Execution time (s) = 1.265490e-04
[joeyma@discovery2 as02]$

```